



# Kubernetes Ingress

# Ingress

Ingress in Kubernetes is an API object that manages external access to services within a cluster, typically HTTP and HTTPS traffic. It allows you to define rules for routing external traffic to services based on various criteria such as the hostname or URL path. This provides a more sophisticated way to manage and expose services compared to using a NodePort or LoadBalancer service type.

# Key Components of Ingress

**1. Ingress Rules:** These are the rules defined within the Ingress resource that determine how incoming traffic should be routed. Rules can be based on host, path, or other criteria

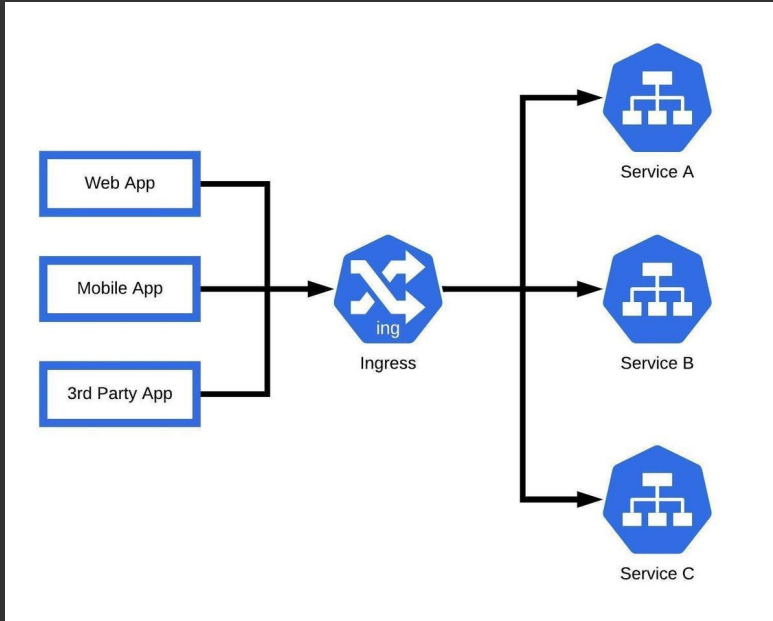
```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: myingress
  labels:
    name: myingress
  annotations:
    kubernetes.io/ingress.class: "gce"
    kubernetes.io/ingress.allow-http: "true"
spec:
  rules:
    - http:
        paths:
          - path: "/"
            pathType: Prefix
            backend:
              service:
                name: api-gateway
                port:
                  number: 80
          - path: "/zipkin"
            pathType: Prefix
            backend:
              service:
                name: zipkin
                port:
                  number: 9411
```

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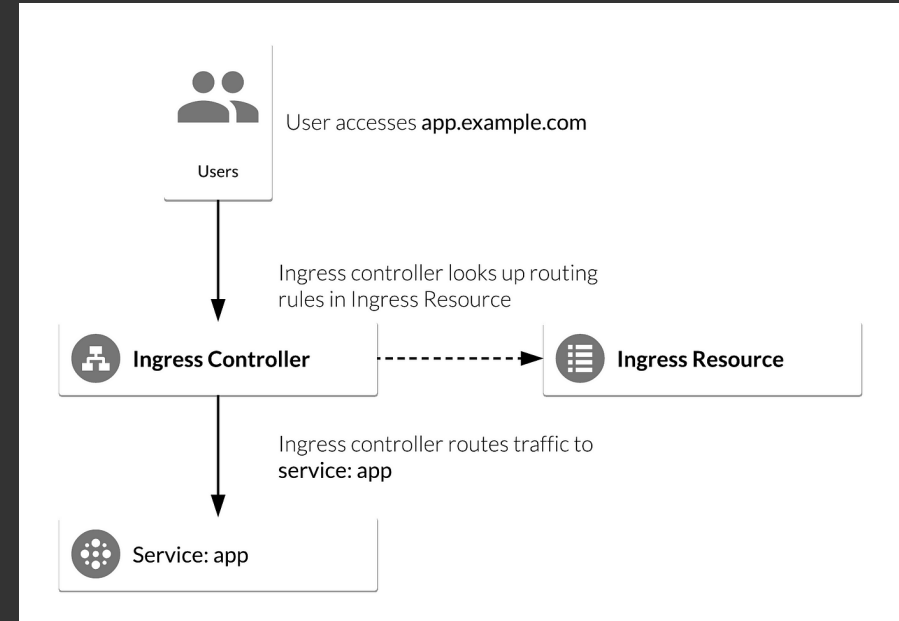
**2. Ingress Resource:** This is a Kubernetes resource that defines the rules for routing incoming traffic. It specifies the hostname, path, and destination services for different types of requests.

**3. Ingress Controller:** This is a specialized load balancer that watches the Ingress resource and processes the rules defined within it. It is responsible for routing incoming traffic to the appropriate services based on the Ingress rules.

# Use cases of Ingress

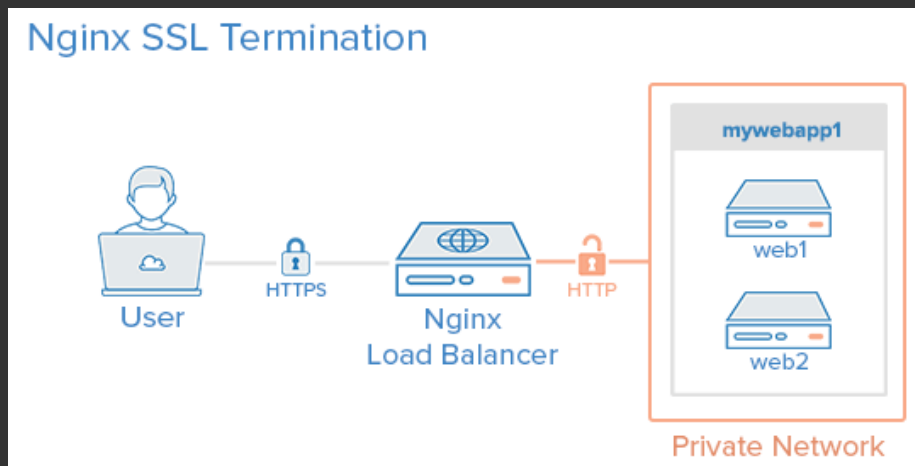


Load Balancing



Domain based Virtual Hosting

## Nginx SSL Termination



SSL Termination

